

Diagnostic Analysis of ATP Matches using Bayesian Networks: Causal Impact of Status and Performance

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Abstract

This study employs Bayesian Networks to model ATP tennis matches (2014-2024), aiming to diagnose the causal weight of pre-match *Status* versus in-match *Performance*. We utilized the `pgmpy` library to implement four architectures, comparing an Expert-defined structure against a Data-Learned one. While the **Learned Model** achieved the highest accuracy (91.51%), the **Expert Model** performed comparably (91.37%) with significantly lower complexity (15 edges vs 27). Our findings suggest that a semantically coherent Expert network is preferable to a denser data-driven one, which is prone to memorizing noise or irrelevant patterns within the data rather than capturing the true underlying causal relationships. By integrating performance metrics, the model serves as a robust diagnostic tool, capable of quantifying the impact of service quality and surface context through counterfactual analysis.

Domain

Introduction

Professional tennis is governed by non-linear dependencies where a player's ranking often does not capture contextual adaptability or momentary form. Unlike deterministic systems, sports outcomes are heavily influenced by stochastic variables that interact in complex ways. This project develops a **Diagnostic Model** to explain the "why" behind match results. By integrating factors known ex-ante (*Status*) with those observed ex-post (Performance statistics like *Aces* and *Service Points*), we create a framework to reason probabilistically about match dynamics, leveraging the Bayesian Network's unique ability to handle uncertainty and causal inference simultaneously.

Aim

The primary objectives of this work are:

- To quantify the causal influence of specific game metrics on the final result, distinguishing between static attributes (Ranking, Age) and dynamic performance variables.
- To evaluate the trade-off between a domain-driven structure (*Expert*) and one automated via *Hill Climbing Search* (*Learned*). The goal is to see if domain knowledge leads to simpler, explainable models that are still accurate.

Method

Data Processing

We analyzed 28,580 ATP matches from the Sackmann repository (1). To ensure the robustness of our diagnostic capabilities and prevent data leakage, we applied a strict temporal split: Training on historical data (2014–2023) and Testing on the 2024 season. Continuous variables (Age, Height, Aces, Service Points) were discretised using **Quantile Binning** (`qcut`), dividing data into three balanced frequency bins ("Low", "Medium", "High") calculated on the training set. This discretization is crucial for Bayesian Networks to effectively model non-linear conditional probabilities.

Geometries

We implemented and compared four Bayesian Network architectures using the `pgmpy` library (2):

1. **Expert Model:** Manually defined causal edges based on domain physics (e.g., *Surface* $\rightarrow$ *Ace Performance*).
2. **Naive Bayes:** A baseline assuming feature independence, used to benchmark the value of modeling dependencies.
3. **Variant:** A simplified Expert model excluding the height node (*Cat.Ht*) to test feature redundancy.
4. **Learned:** Structure discovered via **Hill Climbing Search** utilizing the **BDeu** (Bayesian Dirichlet equivalent uniform) scoring method. This greedy algorithm iteratively adds, deletes, or reverses edges to maximize the network's score, providing a purely data-driven perspective.

Results

The inclusion of ex-post performance variables yields high diagnostic accuracy on the 2024 test set. The data-driven **Learned Model** (91.51%) marginally outperforms the **Expert Model** (91.37%), while Naive Bayes lags (90.01%), confirming that tennis variables are highly dependent and cannot be treated in isolation.

Model Analysis: Expert vs Learned

While accuracy is similar, the structural difference is significant. The **Expert Model** achieves its performance with only **15 edges**, relying on domain-driven assumptions (Figure 1).

Model	Accuracy (2024)
4. Learned (Hill Climbing)	91.51%
1. Expert	91.37%
3. Variant	91.37%
2. Naive Bayes	90.01%

Table 1: Diagnostic Accuracy on Test Set

The **Learned Model** requires **27 edges**, creating a denser network. This comparison highlights a critical insight: automated algorithms tend to identify statistical correlations that may not represent true causal links (spurious correlations), leading to a more complex model without a proportional gain in performance. The Expert structure avoids this overfitting, offering a clearer interpretability of causal chains (e.g., Surface impact) without sacrificing diagnostic power.

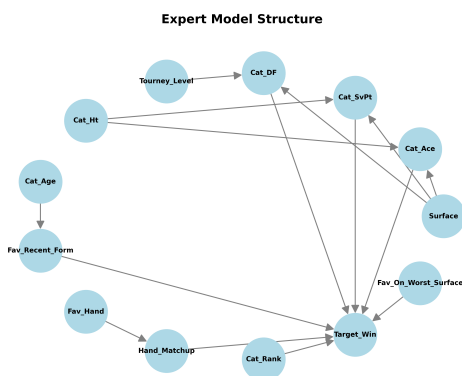


Figure 1: Expert Bayesian Network (Model 1).

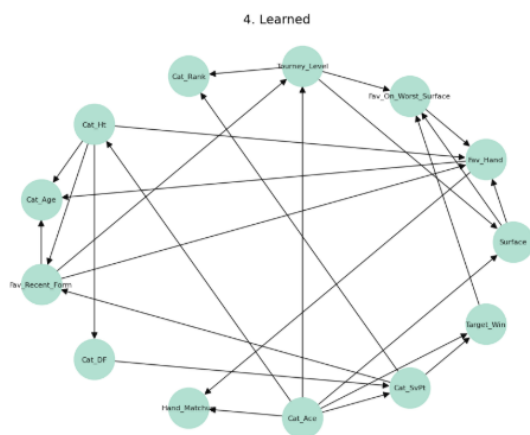


Figure 2: Learned Bayesian Network (Model 4).

Diagnostic Scenarios

We performed inference queries (Variable Elimination) to diagnose the network’s reasoning in specific contexts:

1. The Favorite’s Nightmare (Contextual Vulnerability)

We queried the probability of a ”Strong Favorite” winning when forced to play on their **worst surface**.

- **Result:** The win probability drops to **64.0%** (from a base-line $> 90\%$).
- **Diagnosis:** The model correctly identifies that contextual disadvantage (Surface) can neutralize Ranking superiority.

2. Service Impact (Performance Dominance)

We measured the sensitivity of the outcome to Service Points Won.

- **Result:** The probability swings from **99.8%** (High Service) to **13.2%** (Low Service).
- **Diagnosis:** In-match performance acts as a ”switch” that overrides pre-match status, confirming the model’s diagnostic validity.

3. Pressure Factor (Slam vs ATP 250)

We compared win probabilities for a balanced match in a Grand Slam vs a standard tournament to test stability.

- **Result:** The **Expert Model** shows high stability ($\Delta + 0.1\%$), basing prediction on physical matchups. In contrast, Naive Bayes overestimates the tournament impact ($\Delta + 10.0\%$).
- **Diagnosis:** The Expert model is more robust against ”tournament bias,” focusing on the technical reality of the match rather than the event label.

Conclusion

This project validates the use of Bayesian Networks as a diagnostic tool for tennis. By integrating performance metrics, we transformed the ranking-based prediction problem into a causal analysis of match dynamics. The comparison highlights that a domain-driven structure (**Expert**) is as effective as complex automated search algorithms (**Learned**), offering superior explainability for ”what-if” scenarios. Future work could involve expanding the dataset to include point-by-point data, making the performance diagnosis even more detailed.

Links to external resources

- The notebook containing the project is available on GitHub: https://github.com/sama2908/fundamentals_of_ai_mod3

References

- Sackmann, J. (2024). *ATP Tennis Rankings, Results, and Stats*. GitHub Repository. https://github.com/JeffSackmann/tennis_atp
- Ankan, A., and Panda, A. (2015). *pgmpy: Probabilistic Graphical Models using Python*. Proceedings of the 14th Python in Science Conference.